

Rakibul Islam

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Professional Summary

Enthusiastic, highly motivated, who like to take initiative and seek out new challenges. My research focuses on the intersection of Machine Learning, Deep Learning, and Human-Computer Interaction (HCI). I want to design intelligent and adaptive systems that enhance how humans interact with technology by leveraging data-driven and neural network-based approaches.

Research Areas: Artificial Intelligence, Machine Learning, Computer Vision, Deep Learning, Human-computer interaction

Education

University of Asia Pacific, Bangladesh

Oct 2018–June 2023

B.Sc. in Computer Science and Engineering

Thesis Topic: Deep Learning-Based Background Noise Classification and Reduction for Audio Enhancement.

Relevant Coursework: Structured Programming, Data Structure and Algorithms, Object Oriented Programming, Database, Software Engineering, Machine Learning, Artificial Intelligence, Probability & Statistics, Data Communications, Linear Algebra, Multivariable Calculus, etc.

Thesis

Deep Learning-Based Background Noise Classification and Reduction for Audio Enhancement.

Explanation: I worked on improving the clarity of the audio by reducing background noise. I used an LSTM-based RNN model to detect and reduce background noise in audio. This helped improve the overall clarity and quality of speech..

Professional Experience

Instructor

Oct 2023–Present

Caretutors Technologies Ltd.

Delivered personalized and group-based instruction in Physics, Mathematics, and programming languages (Python, C, C++, JavaScript for Web Development) to high school and college-level students.

Industrial Trainee

Feb 2022–May 2023

Software Engnr. at Brogrammers Lab

Strengthened problem-solving skills by approaching challenges, focusing on effective solutions, and tasks efficiently under deadlines. Improved team collaboration and communication abilities. And developing time management skills in a professional software development environment.

Bdian Youth Research Trainee Program

Sep 2022–Dec 2022

Researcher at Youth Society for Research and Action

Gained hands-on experience in research methodology, including proposing research topics, designing studies, and selecting appropriate methodologies for real-world applications. Developed strong analytical and critical thinking skills through independent research, collaboration with peers, and professional presentations.

Projects

Intent-Explainable Trajectory

PyTorch, ResNet-50, LSTM, LLM (Llama-3/GPT-4)

Designed and implemented an intent-conditioned trajectory prediction system on the PIE dataset. That generates future pedestrian trajectories conditioned on a jointly-trained crossing-intent classifier. Combining a CNN-BiLSTM encoder with an intent-conditioned LSTM decoder. Also integrated an LLM explanation layer that translates model outputs into natural-language justifications for autonomous vehicles.

Object Detection with YOLO

Computer Vision / Deep Learning / Python, PyTorch, OpenCV

Developed as a course project. Built a real-time object detection system using the YOLO architecture to identify and classify multiple objects from images.

Diabetes Prediction System

Machine Learning Application / Python, Pandas, NumPy, Scikit-learn, Logistic Regression

Diabetes is a critical global health issue. A machine learning-based healthcare system that predicts the likelihood of diabetes using patient medical attributes, supporting early diagnosis and preventive care.

Open-Source Personal Website Template for Scholars

Role: Developer / Technologies: HTML, CSS, JavaScript, GitHub

Designed and developed a responsive, modern, and customizable personal website template for professors to showcase research, publications, and teaching. Implemented user-friendly sections, including homepage, about, research, publications, portfolio, blog, and contact.

Medical Insurance Cost Prediction

Regression-Based Machine Learning Project / Linear Regression, Polynomial Regression, Random Forest Regressor

Developed as a course project. A predictive model that estimates medical insurance costs based on demographic and lifestyle factors such as age, BMI, smoking habits, and region.

Technical Skills

Machine Learning and Deep learning algorithms

Programming Languages: Python, JavaScript, C/C++

Packages: PyTorch, OpenCV, Pandas, Matplotlib, Keras, Sci-kit learn, TensorFlow

Web Technologies: HTML, CSS, Bootstrap, JavaScript, Json

Tools: Microsoft Word, Excel, PowerPoint, Latex, Overleaf, Power BI, Git

Project Management: Communication Skills (Written & Verbal), Quantitative Skills, Initiative, Interpersonal Skills, Teamwork, Time Management, etc.

Languages: English (Native) IELTS: 6.5 (L:7, W:6, R:6, S:6), Bengali (Fluent)

Certifications

2024 - Mathematics for Machine Learning and Data Science by Luis Serrano | DeepLearning.AI

Linear Algebra for Machine Learning and Data Science , Calculus for Machine Learning and Data Science , Probability & Statistics for Machine Learning and Data Science

2024 - Machine Learning Specialization by Andrew NG | DeepLearning.AI & Stanford University

Supervised Machine Learning: Regression and Classification , Advanced Learning Algorithms , Unsupervised Learning, Recommenders, Reinforcement Learning

2024 - Learn SQL Basics for Data Science Specialization | University of California, Davis

SQL for Data Science by Sadie St. Lawrence

2024 - IBM Data Analyst Professional Certificate | IBM

Data Visualization with Python by Saishruthi Swaminathan and Dr. Pooja

2024 - Data Science Foundations Specialization | IBM

What is Data Science by Alex Aklson

2024 - Google Data Analytics Professional Certificate | Google

Foundations: Data, Data, Everywhere

2022 - Bdian Research Trainee Certificate

References

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